Ahnaf Shahriar

Email | LinkedIn | Github

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Computer Engineering — *Sept. 2021 – May 2026*

- Recipient of Richard & Elizabeth Madter Entrance Scholarship and President's Scholarship of Distinction
- **Relevant Courses**: Digital Communication Systems, Computer Networks, Computer Architecture, Real Time Operating Systems, Digital Hardware systems(Verilog)

Experience

Undergraduate Research Assistant — May. 2025 – Present
University of Waterloo — *Waterloo, ON*

- **Research**: Conducting Case studies on Static Analysis type checking for Java.
- **Compiler Optimization**: Implementing bytecode storage on Nullness checker for EISOP framework.

Wireless Firmware Developer — Jan. 2025 – Apr. 2025
ON Semiconductor Corporation — *Waterloo, ON*

- **SDK Development**: Integrated *I3C support* and *BLE* profile services into sample code.
- **Wireless Connectivity**: Developed and integrated cutting-edge *LE Audio, HFP, and A2DP* functionality into an all-in-one Application.
- **Wireless Manager**: Modularized and implemented a state machine for power efficiency by *25%*
- **Test Automation**: Automated SDK test cases, reducing manual testing efforts by up to *80%*

IC Design and Verification Intern — Jan. 2024 – May 2024
NXP Semiconductors Canada — *Kanata, ON*

- **IP Design**: Designed multiple IP blocks for NXP's flagship dataplane processing SOCs upto *100Gbps*.
- **Timing Analysis**: Spearheaded critical path improvements for IP to meet *600Mhz* from 400Mhz.
- **Funcitonal Testing**: Designed brand new End-to-End functional tests for *ECC Detection* on Chip.

IC Design Verification Intern — May 2023 – Aug. 2023
NXP Semiconductors Canada — *Kanata, ON*

- **UVM SystemVerilog**: Designed the *IP specific Virtual Sequence* to complete corresponding IP coverage exceeding *95%.*
- **Test Planning**: Created Simulation scenarios for testing features in High speed Dataplane processing.
- **Simulation Testing**: Created Simulation scenarios for testing features in High speed Dataplane processing.

Embedded Software Engineering Intern — Sept. 2022 – Dec. 2022
Synapse Product Development — *Seattle, WA*

- **Prototyping**: Leveraged Zephyr RTOS to create a proof of concept on *NRF52 BLE* device.
- **Python APIs**: Developed company specific lab automation software for equipment from *Agilent, Keysight, NI, Tektronik.*
- **Automation**: Streamlined testing and in house procedures using *Python* and *Bash.*
- **Driver Development**: Designed and implemented *drivers* for the controls of PCB testing Device*(I2C, UART)*

Projects

RISC-V processor: A 5 stage pipelined FPGA processor in Verilog. Designed and tested with Vivado for Zynq-7000.
Stereo System: An FPGA designed in Quartus and programmed in C. Implements stereo playback speed options.
Real Time Executable: A STM32 RTOS implementation capable of Pre-emptive task switching and its very own Malloc.
VHDL Compiler: A Java Compiler for creating combinational VHDL circuits. Using a boolean intermediate representation

Technical Skills

Languages: Python, Java, C/C++, Tcl, Bash scripting, ASM, VHDL, SystemVerilog/Verilog
Tools: Quartus, Git, Linux, Qemu, WireShark, UVM, Matlab, Synopsys DC, Cadence Xcelium, Vivado
Hardware: Oscilloscopes, Logic Analyzer, Multimeters, Spectrum Analyzer
Protocols: TCP/IP, JTAG, Serial, Ethernet, CAN/CAN-FD, LIN, AXI